Asignment5- Bivariate Visualization

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```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(haven)
library(dplyr)
library(ggplot2)


setwd("/Users/senorking/Desktop/Fall2025/Data333") # modify as needed
```

Import Data

```r
students <- read.csv("Students and Social Media.csv")
head(students)
```

```
##   Student_ID Age Gender Academic_Level    Country Avg_Daily_Usage_Hours
## 1          1  19 Female  Undergraduate Bangladesh                   5.2
## 2          2  22   Male       Graduate      India                   2.1
## 3          3  20 Female  Undergraduate        USA                   6.0
## 4          4  18   Male    High School         UK                   3.0
## 5          5  21   Male       Graduate     Canada                   4.5
## 6          6  19 Female  Undergraduate  Australia                   7.2
##   Most_Used_Platform Affects_Academic_Performance Sleep_Hours_Per_Night
## 1          Instagram                          Yes                   6.5
## 2            Twitter                           No                   7.5
## 3             TikTok                          Yes                   5.0
## 4            YouTube                           No                   7.0
## 5           Facebook                          Yes                   6.0
```

```
## 6          Instagram                           Yes                 4.5
##   Mental_Health_Score Relationship_Status Conflicts_Over_Social_Media
## 1                   6     In Relationship                           3
## 2                   8              Single                           0
## 3                   5         Complicated                           4
## 4                   7              Single                           1
## 5                   6     In Relationship                           2
## 6                   4         Complicated                           5
##   Addicted_Score
## 1              8
## 2              3
## 3              9
## 4              4
## 5              7
## 6              9
```

Task 1

For this task, I selected the variables Addiction_Score and Sleep_Hours_Per_Night because there may be a relationship between how addicted a student is to social media and how much they sleep.

Addiction_Score - This variable measures how strongly a student is addicted to social media. It is usually recorded on a numerical scale from 1 to 10. Higher values indicate a stronger addiction. Therefore this is a Numeric variable

Sleep_Hours_Per_Night - This variable measures the average number of hours a student sleeps each night. It is measured in hours, therefore it is a numerical variable.

Task 2

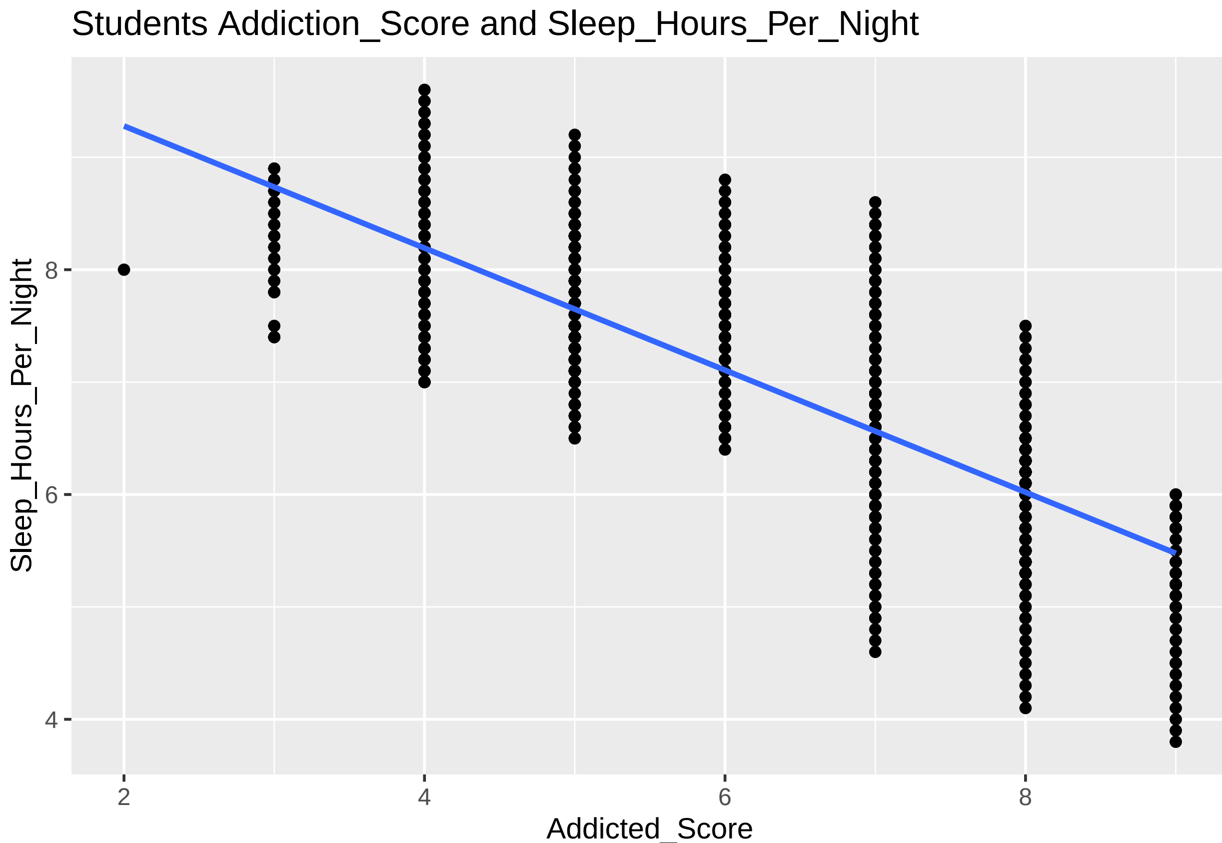
I think Addiction_Score and Sleep_Hours_Per_Night may be correlated because both variables relate to students' daily life. These two variables may be connected because students with higher social media addiction might stay up late using their phone, which could reduce the number of hours they sleep. I think there will be a negative correlation, so as the addiction_score goes up then the hours of sleep students are getting should go down.

Task 3

Visualize the relationship between the two variables Addiction_Score and Sleep_Hours_Per_Night

```r
ggplot(students, aes(x = Addicted_Score, y = Sleep_Hours_Per_Night))+
  geom_point() + geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Students Addiction_Score and Sleep_Hours_Per_Night",
    x = "Addicted_Score",
    y = "Sleep_Hours_Per_Night"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Based on the scatterplot, there is clearly a negative relationship between Addiction_Score and Sleep_Hours_Per_Night. The points generally slope downward, and the trendline shows that students with higher addiction scores reports fewer hours of sleep while students with lower addiction scores tend to sleep more hours on average.

This pattern supports my theory that social media addiction might reduce sleep time.